

Receiver Operating Characteristics

- An **ROC curve** is a graphical representation of a **binary** classifier's performance across different classification thresholds.
 - In general, classifiers compute a score based on some *decision function* (e.g., with Logistic Regression, the decision function is the logistic function).
 - If that score is greater than some threshold, the classifier assigns the input to Class 1, and otherwise to Class 0.
- The ROC curve shows the **True Positive Rate (TPR)** (also known as **Recall** or **Sensitivity**) against the **False Positive Rate (FPR)** at various threshold settings.
- [\[Wiki Reference Link\]](#)

REFERENCE

- Textbook Chapter 3, "Classification"
 - pp 107 - 119
- **True Positive Rate (TPR)**: Measures the proportion of actual positives correctly identified.

$$TPR = \frac{TP}{TP + FN} = \frac{\text{True Positives}}{\text{All Positives}}$$

- **False Positive Rate (FPR)**: Measures the proportion of actual negatives incorrectly classified as positive.

$$FPR = \frac{FP}{FP + TN} = \frac{\text{False Positives}}{\text{All Negatives}}$$

| | **Predicted Positive** | **Predicted Negative** |

|-----|-----|-----| | **Actual Positive** | **True Positive (TP)** | **False Negative (FN)** | | **Actual Negative** | **False Positive (FP)** | **True Negative (TN)** |

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ROC Curve for a Logistic Regression Classifier

Logistic regression outputs probabilities (values between 0 and 1) rather than hard class predictions. These probabilities can be converted to binary predictions by applying a threshold (e.g., 0.5 by default). The ROC curve evaluates the classifier's performance over a range of thresholds.

1. Threshold Adjustment:

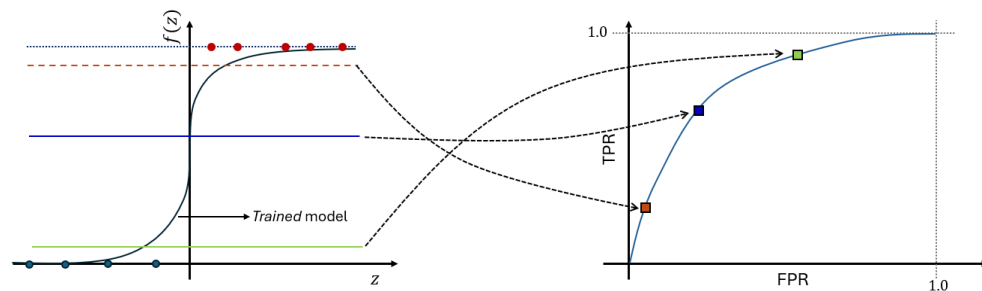
- The threshold determines the cutoff for classifying probabilities into positive or negative classes.
- Lowering the threshold increases both TPR and FPR because more instances are classified as positive.

2. Plotting the ROC Curve:

- **X-axis:** False Positive Rate (FPR).
- **Y-axis:** True Positive Rate (TPR).
- The curve is generated by varying the threshold from 0 to 1 and computing TPR and FPR at each step.

3. Ideal ROC Curve:

- A perfect classifier has an ROC curve that passes through the top-left corner, achieving $TPR = 1$ and $FPR = 0$.
 - A random classifier produces a diagonal line ($FPR = TPR$).
- The following figure illustrates how ROC curves are plotted.
 - On the left: the trained Logistic Regression classifier along with a few training data points, and various thresholds for which TPR and FPR are computed.
 - On the right: the TPR vs FPR curve. The points show (TPR, FPR) values for the various thresholds shown on the right.



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Area Under the Curve (AUC) in ROC Plots

The **AUC** (Area Under the Curve) is a single-number summary of the **Receiver Operating Characteristic (ROC) curve**, which provides a graphical representation of a classifier's performance across various threshold values.

What Does AUC Represent?

The AUC measures the **entire two-dimensional area** underneath the ROC curve. It quantifies the overall ability of the model to distinguish between positive and negative classes, independent of the decision threshold.

- **Range:** AUC values range from 0 to 1.
 - **1.0 (Perfect):** The classifier perfectly distinguishes between positive and negative classes.
 - **0.5 (Random Guessing):** The classifier has no discriminative power (e.g., flipping a coin).
 - **< 0.5:** The classifier performs worse than random guessing, indicating an inverted decision-making process.

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Why adjust thresholds?

- To see why one would wish to use a threshold other than $f(z) = 0.5$, consider the simple case of declaring a defendant "guilty" during a legal trial.
- This is clearly a classification problem where the jury is required to decide whether the prosecution has presented enough evidence to establish the guilt or lack thereof of the defendant.
- Following our nomenclature until now, we will give "guilty" the label '1' and "not guilty" the label '0'. With that, the confusion matrix is of the form | | **Declared Guilty** | **Declared Innocent** |

|-----|-----|-----| | **Actually Guilty** | **True Positive (TP)** | **False Negative (FN)** | | **Actually Innocent** | **False Positive (FP)** | **True Negative (TN)** |

- Given the fact that it would be unconscionable to convict an innocent person, the focus of trials is to minimize false positives.

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- Alternatively, consider the problem of using a medical test to detect an infection.
- The confusion matrix for this binary classification problem is as follows:

	Tested Infectious	Tested No Infectious
Actually Infectious	True Positive (TP)	False Negative (FN)
Actually Not Infectious	False Positive (FP)	True Negative (TN)

- In cases where the treatment is mild (quarantining, for example), and unlikely to adversely affect a healthy individual, it is better to ensure low false positive rates (i.e., keep threshold high).

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